7/11/17 Chlorophyll Sample

Sample ID	Weight Filtered (g) + flask	Weight of flask (g)	Weight of Sample (g)
FL.01	814.9	472.9	342.0
FL.02	734.9	475.2	259.7
FL.03	693.4	475.0	218.4
FL.04	603.2	474.0	129.2
FL.05	871.3	474.1	397.2
FL.07	752.6	474.4	278.2
FL.14	657.9	473.8	184.1
FL.15	712.8	474.0	238.8
FL.42	598.47	473.7	124.8
FL.44	603.2	473.7	129.5
FL.50	597.5	473.7	123.8
FL.53	592.6	474.0	118.6

Witnessed & Understood by me, Date Invented by: Date

Recorded by:

CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:			
Ship Date:		Contact:	
Destination:		Phone #:	
Recipient:		E-Mail:	
Phone #:			

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Sample Description	Container Type	# Containers	Preservative	Analyses Required	Special Instructions for Lab
20170911	11:05:00				2	HouseDI				2	N	TNO2, TNO2-3 TNH4, TRP, TP, TN	
20170907	16:20:00	FL.02	0.1		1	SW				1	N	"	
20170908	10:25:00	FL.03	0.1		1	SW				1	N	"	
20170908	10:50:00	FL.04	0.1		1	SW				1	N	"	
20170907	16:40:00	FL.05	0.1		1	SW				1	N	"	
20170907	18:55:00	FL.01	0.1		1	SW				1	N	"	
20170907	19:10:00	FL.07	0.1		1	SW				1	N	"	
20170908	10:05:00	FL.14	0.1		1	SW				1	N	"	
20170907	19:50:00	FL.15	0.1		1	SW				1	N	"	
20170907	12:27:00	FL.42	0.1		1	SW				1	N	"	
20170907	10:20:00	FL.44	0.1		1	SW				1	N	"	
20170907	12:00:00	FL.50	0.1		1	SW				1	N	"	
20170907	12:50:00	FL.53	0.1		1	SW				1	N	"	
20170907	16:20:00	FL.02	0.1		1	SW					2M HCl	TOC	
20170908	10:25:00	FL.03	0.1		1	SW					2M HCl	TOC	
20170908	10:50:00	FL.04	0.1		1	SW					2M HCl	TOC	
20170907	16:40:00	FL.05	0.1		1	SW					2M HCl	TOC	
20170907	18:55:00	FL.01	0.1		1	SW					2M HCl	TOC	
20170907	19:10:00	FL.07	0.1		1	SW					2M HCl	TOC	
20170908	10:05:00	FL.14	0.1		1	SW					2M HCl	TOC	
20170907	19:50:00	FL.15	0.1		1	SW					2M HCl	TOC	
20170907	12:27:00	FL.42	0.1		1	SW					2M HCl	TOC	
20170907	10:20:00	FL.44	0.1		1	SW					2M HCl	TOC	
20170907	12:00:00	FL.50	0.1		1	SW					2M HCl	TOC	
20170907	12:50:00	FL.53	0.1		1	SW					2M HCl	TOC	
20170911	11:45:00				2	HouseDI					2M HCl	TOC	

Relinquished By:	Date:	Time:	Received By:	Date:	Time:	Remarks:

CHAIN OF CUSTODY FORM			Originator:					
			Ship Date:			Contact:		
			Destination:			Phone #:		
			Recipient:			E-Mail:		
			Phone #:					

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	Depth	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)
20170907	18:50:00	FL.01	0.1	1	SW	UKN	Chloro	342.0
2017 0907	16:20:00	FL.02	0.1	1	SW	UNK	Chloro	259.7
20170908	14:35:00	FL.03	0.1	1	SW	UNK	Chloro	218.4
20170908	10:50:00	FL.04	0.1	1	SW	UNK	Chloro	129.2
20170907	16:40:00	FL.05	0.1	1	SW	UNK	Chloro	397.2
20170907	19:10:00	FL.07	0.1	1	SW	UNK	Chloro	278.2
20170908	10:05:00	FL.14	0.1	1	SW	UNK	Chloro	184.1
20170907	19:50:00	FL.15	0.1	1	SW	UNK	Chloro	238.8
20170907	12:27:00	FL.42	0.1	1	SW	UNK	Chloro	124.8
20170907	10:20:00	FL.44	0.1	1	SW	UNK	Chloro	129.5
20170907	12:00:00	FL.50	0.1	1	SW	UNK	Chloro	123.8
20170907	12:50:00	FL.53	0.1	1	SW	UNK	Chloro	118.6

Project No. _____

onfidential Book No. _____ TITLE _____

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17 Falls Lake Walker, Daly, Balz Rainy in PM

Blue Traps Boat Problems - fuel line?

Site	Lat / Lon	Depth	Time Deployed
SU-44	36.072755 / -78.779388	3.1	~~10:00~~ 10:00
SU-50	36.068130 / 78.785880	3.0	10:35
SU-42	36.065150 / 78.795521	3.0	10:53
SU-53	36.069253 / 78.794898	2.8	11:14
SU-58	36.068450 / 78.776003	4.9	11:42
SU-56	36.072953 / 78.765908	6.7	11:58
SU-52	36.079761 / 78.760131	3.0	12:09
SU-47	36.068455 / 78.768723	6.1	12:29
SU-43	36.061571 / 78.773015	4.8	12:47
SU-54	36.057921 / 78.785218	5.0	13:18
SU-55	36.053115 / 78.761343	6.0	13:28
49	36.051450 / 78.756533	7.8	13:37
SU-57	36.048253 / 78.752533	6.0	13:44
SU-45	36.044041 / 78.758220	6.5	13:54
SU-10	36.039656 / 78.756116	2.5	14:01
SU-08	36.019023 / 78.748441	11.5	14:18
SU-04	36.012185 / 78.760580	3.0	14:32
SU-03	36.032770 / 78.736945	3.6	14:50
SU-14	36.050463 / 78.717040	5.0	15:06
SU-15	36.022511 / 78.714635	13.0	15:17
SU-11	36.019153 / 78.716818	?	15:30
SU-07	36.005433 / 78.688078	25.0	15:44
SU-01	36.002145 / 78.634311	10.0	16:09
SU-13	36.003700 / 78.646359	17.2	16:27
SU-12	35.965476 / 78.633861	12.0	17:16
SU-05	35.970266 / 78.621273	10.0	17:28
SU-02	35.976073 / 78.589140	10.0	17:40
SU-16	35.956906 / 78.590335	20.0	17:57
24	35.941978 / 78.588026	9.0	18:06
21	Not deployed - Boat Problems		

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Witnessed & Understood by me,	Date	Invented by:	Date

From Page No.

Bubble Trap collected 9/7/17

Site	Time Collected	Volume (mL)	Extainer Code	Notes
SU-44	10:20	19	44BA	Stapler moved
50	12:00	25	50BA	
42	12:27	140+75	42BA 42BB 42BC	
53	12:50	25, 5	53BA	
58	13:14	25, 25, 37	58BA 58BB 58BC	
52	13:35	2	NA	Chamber separated from Fan.
56	13:50	25, 10	56BA	
47	14:08	25, 14	47BA	
43	14:20	13	NA	
24	15:50	15	NA	
16	16:05	25, 23, 25, 10	16BA, 16BB, 16BC	
2	16:20	5	NA	
5	16:40	0	NA	
12	16:55	25, 25, 11	12BA 12BB	
13	17:25	25, 1	13BA	
1	18:50	25	1BA	
7	19:10	25, 25, 25, 55	7BA 7BB 7BC	
11	19:40	1	NA	MAM on trap Not collected
15	19:50	3	NA	
14	10:05	16	NA	
3	10:25	19	3BA	
8	10:35	25, 10	8BA	
4	10:50	18	4BA	
10	11:10	25	10BA	
45	11:20	3.5	NA	
57	11:30	25, 25, 25, 140, 140, 20	57BA 57BB 57BC 57BD 57BE	2 not filled Not sure which
49	12:00	25, 25, 16	49BA 49BB 49BC	
55	12:10	9	NA	
54	12:20	5	NA	Funnel probably disconnected

To Page No.

nessed & Understood by me, Date | Invented by: Date | Recorded by:

DISSOLVED GAS AND AIR DATA SHEET

Lake:	FALLS			Date:	9/07/17		
¥Site	Dissolved gas Exetainer codes		Water Vol/ Air Helium Vol	Atmospheric Air Exetainer Codes			BP (mm Hg)
44	44DA	44DB	25/140	44AA	44AB	44AC	
50	50DA	50DB	25/140				
42	42DA	42DB	25/140				
53	53DA	53DB	25/140				
2	2DA	2DB	25/140	2AA	2AB	2AC	
5	5DA	5DB	25/140				
1	1DA	1DB	25/140				
7	7DA	7DB	25/140				
15	15DA	15DB	25/140				
14	14DA	14DB	25/140				
3	3DA	3DB	25/140				
4	4DA	4DB	25/140				

9/8/17

Water Temp
22.6
25.0
24.5
25.5
27.5
25.0
23.9
23.0
25.0
25.2
24.9
24.3

¥Site: all dissolved gas samples collected at 0.1m depth
Boat motor should be off for 2 minutes before sample collection
Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	BAd EiF clock + 4:01:16 relative to LGR clock
Lake:	FALLS
Sample Date:	9/7/17
GGA file	GGA_2017_09_07_F0003
gps file name	BAD EIF

GGA logger Time 12:56:44 BAD EIF Time 16:58:00

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber volume (L)	CH_4 and CO_2 profile checked
44	11:35:30	N	Y BADEIF	55	Yes
50	~~11:40:40~~	N	continued	55	Yes
42	12:35:32	N	continued	55	Yes
53	12:54:53	N	continued	55	Yes
58	13:20:41	N	"	~~55~~ 50	Yes
52	13:40:15	N	"	45	Yes
56	13:55:55	N	"	45	Yes
47	14:07:22	N	"	45	Yes
43	14:~~20~~:22 ~~00~~:21	N	"	40	Yes
24	15:51:42	N	"	40	Yes
16	16:05:37	N	"	40	Yes
2	16:26:40	N	"	50	"
5	16:45:40	N	"	45	"
12	17:02:30	N	"	40	"
13	17:28:50				
~~7~~					

12:07:10 start

SONDE DATA SHEET

Lake: FAIS Sample Date: 9/7/17

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond ($\mu s\ cm^{-1}$)	pH	Chl a (ug/L)	Turbidity (NTU)	ORP
SV-44	~~3.5~~	0.1	23.21	7.99	93.5	0.172	8.41	16.1	48.0	142.9
50	3.1	0.1	24.65	9.66	116.1	0.187	7.76	17.7	40.2	115.8
42	3.0	0.1	24.46	9.75	116.8	0.241	7.79	16.2	36.9	103.8
53	3.0	0.1	24.80	8.66	104.6	0.197	7.81	16.9	49.4	105.8
58	4.9	0.1	24.62	8.70	104.6	0.165	7.91	16.5	48.2	107.5
52	3.0	0.1	26.18	11.65	144.6	0.142	8.43	14.2	22.7	71.6
56	6.8	0.1	25.82	9.76	120.0	0.145	8.14	11.6	25.3	96.3
47	6.5	0.1	25.14	10.00	121.2	0.146	8.13	14.1	25.4	
43	4.1	0.1	24.71	10.18	122.5	0.151	8.07	13.7	25.0	
24	8.0ft	0.1	28.03	11.35	145.0	0.083	8.18	4.6	4.5	
16	20	0.1	27.99	7.57	96.9	0.083	8.01	5.8	2.6	
2	11	0.1	28.00	10.11	129.3	0.083	8.22	4.7	8.2	
5	12	0.1	27.80	7.22	92.1	0.082	8.10	6.4	2.8	
12	11	0.1	27.52	9.10	109.1	0.049	8.17	6.2	1.2	
13	16.6	0.1	27.98	10.01	125.2	0.081	8.08	11.3	3.1	
1	11	0.1	27.30	10.89	137.5	0.080	7.64	13.0	3.1	
7	23.8	0.1	26.29	9.95	123.2	0.079	7.68	14.7	4.5	
7	23.8	20ft	25.42	4.90	59.8	0.080	7.72	9.5	11.6	
7	~~10~~ 23.8	10 ft	26.11	7.95	97.5	0.079	7.59	13.1	4.8	
11	9.5	0.1	25.43	10.14	124.9	0.082	7.90	11.4	10.8	
15	13.0	0.1	25.40	9.84	120.1	0.082	7.81	12.0	9.7	
14	5.0	0.1	24.61	8.2	98.5	0.080	7.99	9.6	13.7	148.2
3	3.5	0.1	23.4	9.38	110.0	0.087	7.86	6.9	10.0	134
~~8~~ 4	2.8	0.1	23.85	9.49	111.0	0.087	7.82	14.4	21.9	126.6
~~8~~ 8	11.5	0.1	25.06	9.61	116.8	0.091	7.69	16.2	14.8	120.7
10	3.0	0.1	24.62	9.69	116.3	0.096	7.64	9.5	14.5	120.4
45	7.0	0.1	24.95	9.39	112.6	0.098	7.73	~~16.4~~	15.0	124.5
57	6.0	0.1	25.26	10.37	127.1	0.126	7.67	10.6	19.1	118.9
57	6.0	0.1	23.88	0.167	6.2	0.017	7.11	0?	700?	14.3
49	8.0	0.1	23.81	8.341	98.3	0.133	7.32	10.9	78.0	87.5
Signatu				Date				~~15.1~~ 14.0		
55	6.0	0.1	24.31	9.57	113.4	0.135	7.48		26.1	79.9
54	5.5	0.1	24.43	10.41	124.4	0.150	7.69	12.8	26.0	74.5

9/8/17

Post check w/ chloro & Turbidity

Cals

DOX.

Dry	88.7 % DO
Cal	99.6 % DO

pH 7.0

Dry	~~6.95~~ 7.07 pH
Cal	6.99 pH

pH 11.0

Dry	10.66 pH
Cal	11.01 pH

Turbidity

Zero	-0.1 NTU
Cal	0.0 NTU

Chlorophyll

Zero	0.4 - 1.0 mg/L
Cal	0.2 mg/L

ORP ~ Uncalibrated

Conductivity ~ Uncalibrated